# *Nutri Ease*

*Smart Bites, Better Lives*

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### *Problem Statement*

Many individuals struggle with planning their meals due to time constraints, dietary restrictions, and budget considerations. Our platform aims to simplify meal planning by offering personalized 7-day meal plans that are healthy, budget-friendly, and nutritious.

### *Why We Chose This Problem*

Effective meal planning is essential for maintaining a healthy lifestyle. However, existing solutions often lack customization and accurate cost estimates. Our platform addresses these gaps by providing tailored meal plans that consider individual user preferences and constraints.

### *Our Approach*

We developed a web-based Meal Planner that collects user inputs such as:

- Weight goals
- Activity levels
- Dietary restrictions (allergies, vegetarian, vegan, etc.)
- Cultural preferences
- Age
- Height

Using this information, our platform generates a 7-day meal plan that includes:

- Cost per serving estimates
- List of required ingredients with quantities
- Approximate pricing per serving

Additionally, users can input available ingredients from their pantry or refrigerator, allowing the AI model to suggest meals that utilize these items, promoting cost savings and reducing food waste.

### *Technical Stack*

- **Frontend:** Next.js, Tailwind CSS
- **Backend:** Firebase (initially), transitioning to MongoDB for scalability
- **AI Models:** Gemini AI, OpenAI
- **Security:** Cloudflare, Azure Security

### *API Documentation*

- **User Authentication:** Secure login via Firebase Authentication
- **Meal Plan Generation:** AI-powered API generating personalized 7-day meal plans
- **Ingredient Pricing API:** Fetches estimated ingredient prices from a database
- **Nutritional Information API:** Provides detailed nutritional breakdown per meal

### *Deployment*

The Meal Planner is accessible from any device with an internet connection, requiring no installation or setup.

### *Usage Guide*

1. **Home Page:** Users land on the homepage.
2. **User Authentication:** Users create an account or log in via Firebase Authentication.
3. **Form Submission:** Users fill out a form with personal dietary details.
4. **Meal Plan Generation:** A customized 7-day meal plan is generated based on the provided details.
5. **Review & Adjustments:** Users can review the meal plan and adjust preferences if needed.
6. **Ingredient & Cost Breakdown:** The platform provides a breakdown of required ingredients, their quantities, and estimated pricing.

### *Security Considerations*

- **Cloudflare:** Protects against DDoS attacks and enhances website performance.
- **Azure Security Services:** Monitors and secures user data to prevent breaches.
- **Secure API Calls:** Data encryption to prevent unauthorized access.

### *Achievements*

- **Microsoft Startup Recognition:** Our project has been recognized by Microsoft's startup program, providing us with resources and support to enhance our platform.
- **Funding:** We have secured funding from Microsoft to further develop and scale our solution.

### *Alignment with Sustainable Development Goals (SDGs)*

Our project contributes to the following United Nations Sustainable Development Goals:

- **Goal 2: Zero Hunger:** By providing personalized meal plans that consider nutritional needs and budget constraints, we aim to improve food security and promote sustainable agriculture.
- **Goal 3: Good Health and Well-being:** Our platform encourages healthy eating habits, contributing to overall well-being and the prevention of diet-related diseases.
- **Goal 12: Responsible Consumption and Production:** By suggesting meals based on available ingredients, we help reduce food waste and promote efficient use of resources.

### *Future Improvements*

- **Real-time Ingredient Price Tracking:** Integrate with grocery stores to provide up-to-date pricing.
- **Fitness App Integration:** Sync with fitness applications for a holistic health approach.
- **AI-based Recipe Suggestions:** Offer recipes based on user preferences and available ingredients.
- **Community Features:** Enable users to share meal plans and recipes for enhanced engagement.